

PROCESS OF NORMALIZATION:

The following analysis of normalization evaluates five key tables:

- AccountType
- Account
- Invoice
- JournalEntry
- JournalItem

1- FIRST NORMAL FORM (1NF):

Rule:

All attributes must be atomic, each row is uniquely identifiable via Primary key (PK), and no repeating groups.

- **AccountType:**

It satisfies 1NF, typeId is an atomic PK and categoryName is a single string value.

- **Account:**

It satisfies 1NF because accountId provides unique identification, and attributes like balance are atomic single values.

- **Invoice:**

Satisfies 1NF, invoiceId is a unique PK and it contains no multi-valued attributes or lists.

- **JournalEntry:**

Satisfies 1NF, entryId is PK. Transaction lines are stored in a separate table to avoid repeating groups within the table.

- **JournalItem:**

Satisfies 1NF, the composite key (entryId, lineNo) uniquely identifies every row. Amounts are atomic decimals.

CONCLUSION:

The schema satisfies 1NF because every table features a uniquely identifying primary key and strictly atomic attributes. Repeating groups are eliminated by design, establishing a structurally sound relational baseline.

2- **SECOND NORMAL FORM (2NF):**

Rule:

Must be in 1NF and contain no partial dependencies; non-key attributes must depend on the whole

Primary key.

- **AccountType, Account, Invoice, JournalEntry:**

These tables satisfy 2NF automatically because they use single column Primary keys, making partial dependency logically impossible.

• **JournalItem:**

Satisfies 2NF. All attributes, such as debit, credit and accountId, depend on the full composite PK (entryId+lineNo). Attributes depending only on entryId (e.g. date) are correctly located in JournalEntry table.

CONCLUSION:

The database fulfills 2NF standards by completely eliminating partial functional dependencies across all entities. Most importantly, the JournalItem table ensures every financial attribute relies entirely on full composite Primary key.

3- **SETHIRD NORMAL FORM(3NF):**

Rule:

Must be in 2NF and contain no transitive dependencies; non-key attributes must not depend on other non-key attributes.

AccountType:

Satisfies 3NF; there is only one non-key attribute, preventing transitive dependencies.

Account:

Satisfies 3NF, categoryName is excluded and accessed via Foreign Key (FK) join to accountType, breaking the transitive chain.

- **Invoice:**

Satisfies 3NF, descriptive data like customerName is not stored here, preventing dependency on the customerId non-key attribute.

- **JournalEntry:**

Satisfies 3NF, all descriptive data for employees or vendors is kept in their respective tables.

- **JournalItem:**

Satisfies 3NF, accountName and taxPercentage are accessed via JOINs rather than being duplicated in this table.

CONCLUSION:

The transition to 3NF successfully eradicates all transitive dependencies by offloading descriptive data to independent tables. The strategic use of Foreign Keys prevents data redundancy and protects the system against update anomalies.

4- BOYCE-CODD NORMAL FORM:

Rule:

Must be in 3NF and for every functional dependency $(X \rightarrow Y)$, X must be a super key.

- **AccountType, Account, Invoice, JournalEntry:**

In these tables, the only determinants are the Primary Keys, which are all Super Keys.

- **JournalItem:**

Satisfies BCNF. The Full composite PK (entryId, lineNo) is the only determinant for the other functional attributes in the table.

CONCLUSION:

The schema achieves BCNF because every functional determinant across all five tables operates as a valid candidate superkey. This optimal structure guarantees the absence of anomalies, meaning no further decomposition is required.